

GUAYAQUIL, Ecuador

WMO: 842030

Lat: 2.157S

Lon: 79.884W

Elev: 6

StdP: 101.26

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.9	19.2	15.0	10.7	22.3	15.8	11.2	22.2	8.2	22.6	7.7	22.7	3.9	230	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	6.7	32.9	24.0	32.1	24.1	31.8	24.1	26.6	30.0	26.0	29.5	25.5	29.2	2.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.1	28.8	25.0	20.1	27.9	24.2	19.1	27.5	83.3	30.1	80.4	29.8	78.3	28.9	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
7.3	6.6	6.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			18.7	34.5	1.9	0.8	17.3	35.1	16.2	35.6	15.2	36.1	13.8	36.7
			15.8	26.5	1.9	0.8	14.4	27.1	13.3	27.6	12.3	28.1	10.9	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	26.9	26.8	27.3	27.2	26.8	25.3	24.7	24.7	25.2	25.2	25.7	26.9
	DBStd	1.66	1.41	1.33	1.23	1.17	1.32	1.50	1.54	1.41	1.43	1.22	1.25	1.37
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5859	524	470	536	517	519	460	456	457	456	471	470	524
	CDD18.3	2818	266	237	277	267	261	210	198	198	206	212	220	266
	CDH23.3	22402	2381	2097	2659	2606	2349	1476	1217	1186	1333	1289	1497	2312
	CDH26.7	6636	739	577	828	846	732	369	272	286	380	339	440	827
(14) Wind	WSAvg	2.9	2.2	1.9	2.0	2.2	2.5	3.0	3.4	3.6	3.7	3.7	3.4	3.1
(15) Precipitation	PrecAvg	866	200	249	276	159	51	3	0	1	2	2	1	33
	PrecMax	2061	701	753	911	604	208	28	8	20	21	18	6	294
	PrecMin	329	14	44	7	11	0	0	0	0	0	0	0	0
	PrecStd	439	158	180	203	139	68	6	1	3	5	4	1	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	32.8	32.9	33.0	32.7	31.9	31.8	31.5	32.2	32.1	32.2	33.9
		MCWB	23.2	25.5	25.5	24.8	24.6	24.0	23.9	23.9	23.7	23.1	23.3	23.3
	2%	DB	32.2	31.8	32.0	32.0	31.8	30.9	30.2	30.2	31.0	30.8	31.1	32.8
		MCWB	23.7	25.0	25.1	24.7	24.7	23.6	23.5	23.2	23.3	22.9	23.1	23.1
	5%	DB	31.2	30.9	31.1	31.2	31.0	29.9	29.1	29.2	30.0	29.8	30.1	31.8
		MCWB	23.7	24.7	24.8	24.6	24.4	23.3	22.9	22.7	22.9	22.5	22.7	23.0
	10%	DB	30.2	29.9	30.2	30.7	30.1	28.9	28.1	28.2	29.0	28.8	29.1	30.8
		MCWB	23.5	24.5	24.6	24.5	24.2	23.0	22.4	22.2	22.4	22.1	22.2	22.8
	0.4%	WB	27.3	27.8	27.8	26.5	26.2	25.1	25.2	25.0	25.2	24.0	26.2	24.9
		MCDB	30.5	31.1	31.2	30.9	30.5	29.9	30.4	30.4	30.4	30.5	29.8	30.8
	2%	WB	25.7	26.3	26.4	25.7	25.7	24.3	24.2	23.9	23.9	23.3	24.5	24.2
		MCDB	28.4	29.2	29.8	30.1	29.9	29.1	29.3	29.3	29.7	29.4	28.3	30.1
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	24.9	25.7	25.8	25.2	25.1	23.7	23.4	23.1	23.2	22.7	23.5	23.7
		MCDB	28.4	28.6	29.2	29.6	29.4	28.4	28.0	28.3	28.8	28.5	28.4	29.5
	10%	WB	24.3	25.1	25.2	24.8	24.6	23.2	22.7	22.5	22.6	22.3	22.7	23.2
		MCDB	28.1	28.1	28.8	29.1	28.8	27.8	27.2	27.5	28.1	27.8	28.2	29.0
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.0	6.3	6.7	6.6	6.8	6.9	7.5	8.0	7.6	7.7	7.8
		MCDBR	8.0	7.1	7.2	7.5	7.6	8.0	8.0	8.6	9.1	9.1	9.0	9.2
		MCWBR	3.0	2.9	3.1	3.5	3.5	3.6	3.6	4.0	4.0	4.0	3.8	3.5
	5% WB	MCDBR	6.5	6.2	6.6	6.9	6.8	7.2	7.4	8.0	8.4	8.2	8.0	7.9
		MCWBR	2.9	2.9	3.1	3.3	3.5	3.5	3.5	3.9	3.9	3.8	3.6	3.3
(40) Clear-Sky Solar Irradiance	taub		0.507	0.505	0.502	0.553	0.556	0.525	0.507	0.543	0.588	0.549	0.566	0.555
	taud		2.157	2.179	2.192	2.027	2.007	2.100	2.162	2.049	1.921	2.033	1.965	2.002
	Ebn at Noon		834	840	834	770	741	753	775	768	754	793	778	787
	Edh at Noon		159	157	154	176	173	154	146	170	199	179	191	184
(44) All-Sky Solar Radiation	RadAvg		4.11	4.16	4.48	4.56	4.09	3.65	3.58	3.82	4.13	4.03	4.24	4.25
	RadStd		0.42	0.23	0.33	0.25	0.34	0.30	0.30	0.34	0.29	0.33	0.42	0.34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page